

Python Programming Exam

Questions

[Time: 1 hr]
[Total Marks: 100]

	Perform the following tasks:	Marks								
Q1.	Count all letters, digits, and special symbols from a given string str1 = "I@#ma26rt^&ic5us"	[5]								
Q2.	Write a Python program to find numbers divisible by nineteen or thirteen from a list of numbers using Lambda. a=[19, 65, 57, 39, 152, 639, 121, 44, 90, 190]									
Q3.	Find all of the words in a string that are less than 5 letters string = "Practice Problems to Drill List Comprehension in Your Head."									
Q4.	Calculate income tax for the given income by adhering to the below rules <table><thead><tr><th>Taxable Income</th><th>Rate (in %)</th></tr></thead><tbody><tr><td>First \$10,000</td><td>0</td></tr><tr><td>Next \$10,000</td><td>10</td></tr><tr><td>The remaining</td><td>20</td></tr></tbody></table> Expected Output: For example, suppose the taxable income is 45000 the income tax payable is 10000 0% + 10000 10% + 25000 20% = \$6000.	Taxable Income	Rate (in %)	First \$10,000	0	Next \$10,000	10	The remaining	20	
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Q5.	Given below are the list of positive and negative words. Also a list of tweets is provided. Separate out the positive and negative comments¶ <pre>positive = ['good','awesome', 'best', 'nice'] negative = ['worst','awful', 'bad']</pre> <pre>tweets = ['This government policies are good', 'bad implementation', 'The way he played showed that he is one of the best players in the world', 'Her acting in the play was awesome', 'The wine tastes awful', 'It's nice to hear this little kid's laugh']</pre>	
Q6.	Extend nested list by adding the sublist You have been given a nested list. Write a program to extend it by adding the sublist ["h", "i", "j"] in such a way that it will look like the following list. <pre>list1 = ["a", "b", ["c", ["d", "e", ["f", "g"], "k"], "l"], "m", "n"]</pre> sub list to add <pre>sub_list = ["h", "i", "j"]</pre> expected output =['a', 'b', ['c', ['d', 'e', ['f', 'g', 'h', 'i', 'j'], 'k'], 'l'], 'm', 'n'] Remove and add item in a list Write a program to remove the item present at index 4 and add it to the 2nd position and at the end of the list. <pre>list1 = [54, 44, 27, 79, 91, 41]</pre>	
Q7.	Load the iris dataset. Find only those records whose sepal_length = 5.1 and sepal_width = 3.5	

	Check for Maximum petal_length and convert it in to 7.2 and min petal_length and convert it in to 1.2																																														
Q.8	Perform right join to combine values based on the 'ID' in the two dataframes Use the dataframe given below, Read the DataFrame with the help of clipboard function. <table> <thead> <tr> <th>ID</th><th>Candidate_Name</th><th>Subject</th></tr> </thead> <tbody> <tr><td>101</td><td>Alex</td><td>History</td></tr> <tr><td>102</td><td>Amy</td><td>English</td></tr> <tr><td>103</td><td>Allen</td><td>Geography</td></tr> <tr><td>104</td><td>Alice</td><td>German</td></tr> <tr><td>105</td><td>James</td><td>History</td></tr> <tr><td>106</td><td>Sara</td><td>German</td></tr> <tr><td>107</td><td>Mia</td><td>English</td></tr> </tbody> </table> <table> <thead> <tr> <th>ID</th><th>City</th><th>Subject_Score</th></tr> </thead> <tbody> <tr><td>101</td><td>Delhi</td><td>89</td></tr> <tr><td>102</td><td>Mumbai</td><td>78</td></tr> <tr><td>103</td><td>Delhi</td><td>77</td></tr> <tr><td>104</td><td>Chennai</td><td>87</td></tr> <tr><td>105</td><td>Hyderabad</td><td>87</td></tr> <tr><td>108</td><td>Delhi</td><td>84</td></tr> </tbody> </table>	ID	Candidate_Name	Subject	101	Alex	History	102	Amy	English	103	Allen	Geography	104	Alice	German	105	James	History	106	Sara	German	107	Mia	English	ID	City	Subject_Score	101	Delhi	89	102	Mumbai	78	103	Delhi	77	104	Chennai	87	105	Hyderabad	87	108	Delhi	84	
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